

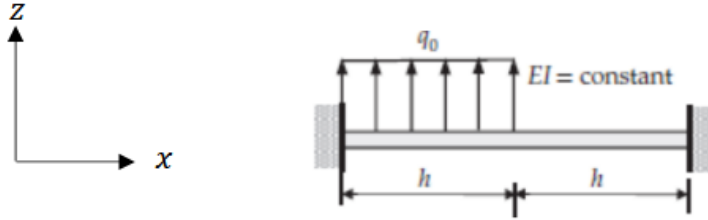
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Intro to Finite Element Method (CIV E – 665)
 Assignment No. 5



1 Question 1

Use **two beam elements** to solve for the transverse displacement $W(x)$ of the following beam.



1.1 I. Find the element wise stiffness matrix and force vectors.

The stiffness matrix and force vectors will have the following form:

$$K^e = \frac{2E^e I^e}{L_e^3} \begin{bmatrix} 6 & -3L_e & -6 & -3L_e \\ -3L_e & 2L_e^2 & 3L_e & L_e^2 \\ -6 & 3L_e & 6 & 3L_e \\ -3L_e & L_e^2 & 3L_e & 2L_e^2 \end{bmatrix}$$

$$f^e = \left\{ \int_{x_1^e}^{x_2^e} \psi_i^e q dx \right\}$$

Where:

- $L_e = h$

For this problem, due to $L_1 = L_2$, $EI = \text{constant}$, implies $K_1 = K_2$, so,

$$K^1 = K^2 = \frac{2EI}{h^3} \begin{bmatrix} 6 & -3h & -6 & -3h \\ -3h & 2h^2 & 3h & h^2 \\ -6 & 3h & 6 & 3h \\ -3h & h^2 & 3h & 2h^2 \end{bmatrix} \quad (1)$$

Since q_0 is a uniform distributed load, we can use,

$$f^e = \frac{q^e L_e}{12} \begin{Bmatrix} 6 \\ -L_e \\ 6 \\ L_e \end{Bmatrix}$$

NB: This is the positive sign convention adopted by J. N. Reddy

- Up: Positive • Clockwise: Positive

For element 1, $q^e = q_0$

$$f^e = \frac{q_0 h}{12} \begin{Bmatrix} 6 \\ -h \\ 6 \\ h \end{Bmatrix} \quad (2)$$

For element 2, $q^e = 0$

$$f^e = \frac{q_0 h}{12} \begin{Bmatrix} 6 \\ -h \\ 6 \\ h \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (3)$$

1.2 II. Assemble the system.

For this beam, we use two beam elements to discretize and assemble it.

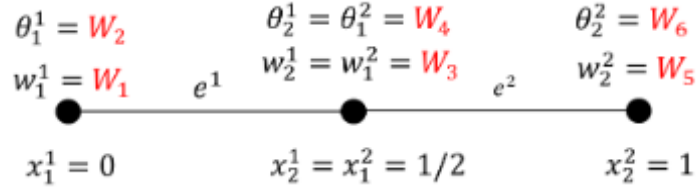


Figure 1: Symbol numbering system

Each beam element has 4 degrees of freedom. Assign a coordinate system as follows:

- x_n^e : Position of node n for element e
- w_n^e : Local Transverse displacement of node n for element e
- θ_n : Local Slope of node n for element e
- W : Global degree of freedom

For element 1,

$$K^1 = \begin{bmatrix} K_{11}^1 & K_{12}^1 & K_{13}^1 & K_{14}^1 \\ K_{21}^1 & K_{22}^1 & K_{23}^1 & K_{24}^1 \\ K_{31}^1 & K_{32}^1 & K_{33}^1 & K_{34}^1 \\ K_{41}^1 & K_{42}^1 & K_{43}^1 & K_{44}^1 \end{bmatrix}, \quad f^1 = \begin{Bmatrix} f_1^1 \\ f_2^1 \\ f_3^1 \\ f_4^1 \end{Bmatrix}$$

For element 2,

$$K^2 = \begin{bmatrix} K_{11}^2 & K_{12}^2 & K_{13}^2 & K_{14}^2 \\ K_{21}^2 & K_{22}^2 & K_{23}^2 & K_{24}^2 \\ K_{31}^2 & K_{32}^2 & K_{33}^2 & K_{34}^2 \\ K_{41}^2 & K_{42}^2 & K_{43}^2 & K_{44}^2 \end{bmatrix}, \quad f^2 = \begin{Bmatrix} f_1^2 \\ f_2^2 \\ f_3^2 \\ f_4^2 \end{Bmatrix}$$

Since this system has 6 independent degrees of freedom, the global system will be:

$$\{K\}_{6 \times 6} \{W\}_{6 \times 6} = \{f\}_{6 \times 6}$$

Assemble Equation 1 and Equation 2,

$$\begin{bmatrix} K_{11}^1 & K_{12}^1 & K_{13}^1 & K_{14}^1 & 0 & 0 \\ K_{21}^1 & K_{22}^1 & K_{23}^1 & K_{24}^1 & 0 & 0 \\ K_{31}^1 & K_{32}^1 & K_{33}^1 + K_{11}^2 & K_{34}^1 + K_{12}^2 & K_{13}^2 & K_{14}^2 \\ K_{41}^1 & K_{42}^1 & K_{43}^1 + K_{21}^2 & K_{44}^1 + K_{22}^2 & K_{23}^2 & K_{24}^2 \\ 0 & 0 & K_{31}^2 & K_{32}^2 & K_{33}^2 & K_{34}^2 \\ 0 & 0 & K_{41}^2 & K_{42}^2 & K_{43}^2 & K_{44}^2 \end{bmatrix} \begin{Bmatrix} W_1 \\ W_2 \\ W_3 \\ W_4 \\ W_5 \\ W_6 \end{Bmatrix} = \begin{Bmatrix} f_1^1 \\ f_2^1 \\ f_3^1 + f_1^2 \\ f_4^1 + f_2^2 \\ f_3^2 \\ f_4^2 \end{Bmatrix} \quad (4)$$

Substituting Equation 1, Equation 2, and Equation 3 into Equation 4, we have,

$$\begin{bmatrix} \frac{12EI}{h^3} & -\frac{6EI}{h^2} & -\frac{12EI}{h^3} & -\frac{6EI}{h^2} & 0 & 0 \\ -\frac{6EI}{h^2} & \frac{4EI}{h} & \frac{6EI}{h^2} & \frac{2EI}{h} & 0 & 0 \\ \frac{12EI}{h^3} & \frac{6EI}{h^2} & \frac{24EI}{h^3} & 0 & -\frac{12EI}{h^3} & -\frac{6EI}{h^2} \\ -\frac{6EI}{h^2} & \frac{2EI}{h} & 0 & \frac{8EI}{h} & \frac{6EI}{h^2} & \frac{2EI}{h} \\ 0 & 0 & -\frac{12EI}{h^3} & \frac{6EI}{h^2} & \frac{12EI}{h^3} & \frac{6EI}{h^2} \\ 0 & 0 & -\frac{6EI}{h^2} & \frac{2EI}{h} & \frac{6EI}{h^2} & \frac{4EI}{h} \end{bmatrix} \begin{Bmatrix} W_1 \\ W_2 \\ W_3 \\ W_4 \\ W_5 \\ W_6 \end{Bmatrix} = \begin{Bmatrix} \frac{q_0 h}{2} \\ -\frac{q_0 h^2}{12} \\ \frac{q_0 h}{2} \\ \frac{q_0 h^2}{12} \\ 0 \\ 0 \end{Bmatrix}$$

1.3 III. Implement the boundary conditions.

In this problem, the beam is fixed at both ends, so,

$$\begin{aligned} w_1^1 &= w_2^2 = 0 \\ \theta_1^1 &= \theta_2^2 = 0 \end{aligned}$$

Per the relationship of the Figure 1, $W_1 = W_2 = W_5 = W_6 = 0$. Set the elements of the corresponding row and f are 0. Equation 5 becomes as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{12EI}{h^3} & \frac{6EI}{h^2} & \frac{24EI}{h^3} & 0 & -\frac{12EI}{h^3} & -\frac{6EI}{h^2} \\ -\frac{6EI}{h^2} & \frac{2EI}{h} & 0 & \frac{8EI}{h} & \frac{6EI}{h^2} & \frac{2EI}{h} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} W_1 \\ W_2 \\ W_3 \\ W_4 \\ W_5 \\ W_6 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \frac{q_0 h}{2} \\ \frac{q_0 h^2}{12} \\ 0 \\ 0 \end{Bmatrix} \quad (5)$$

1.4 IV. Solve for the nodal values and plot $W(x)$

Given: $h = 1m$ $q_0 = 100N/m$ $EI = 10^6 Nm^2$

Use Matlab to solve Equation 5. The source code is as follows:

```
W =  
  
1.0e-05 *  
  
0.2083  
0.1042
```

To construct the full solution:

$$w^e(x) = \sum_{j=1}^4 W_j \psi_j(x)$$

For element 1,

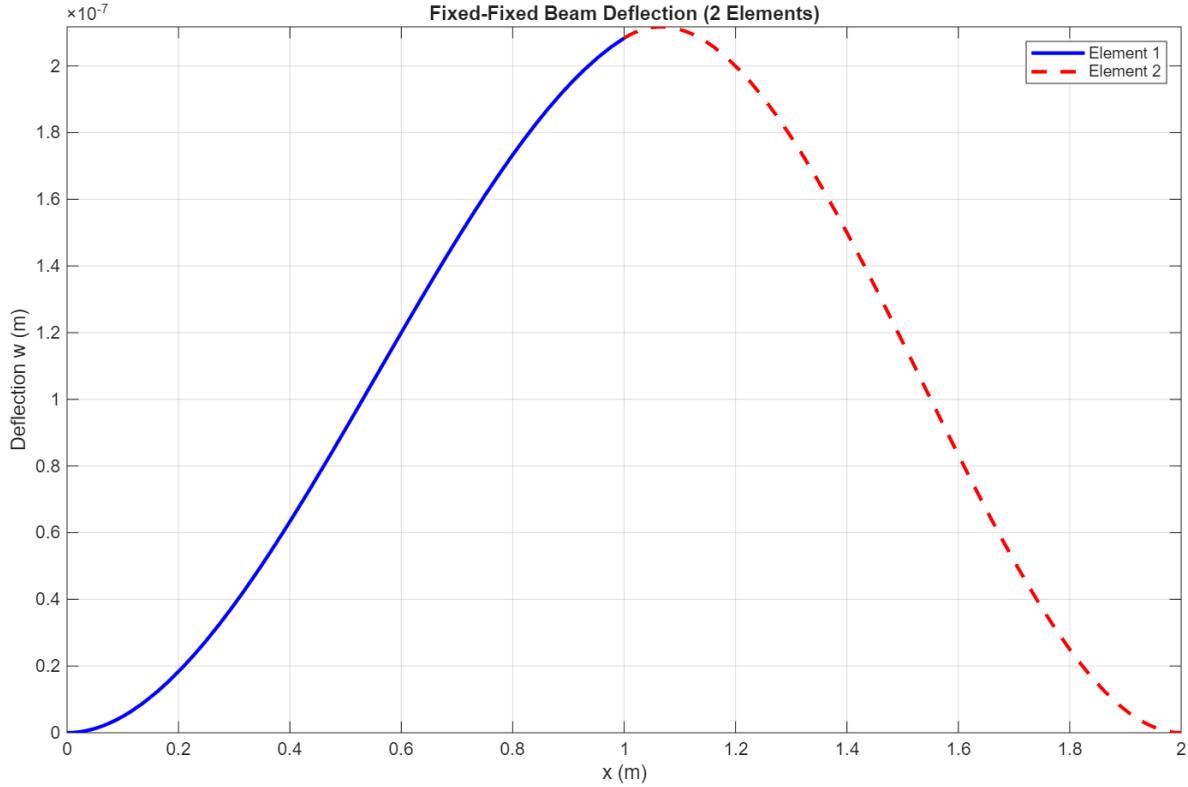
$$\begin{aligned}
 w^1(x) &= \sum_{j=1}^4 W_j \psi_j(x) \\
 &= 0 \times \psi_1 + 0 \times \psi_2 + W_3 \psi_3 + W_4 \psi_4 \\
 &= 2.083 \times 10^{-6} (3x^2 - 2x^3) + 1.042 \times 10^{-6} (-x^2 + x^3)
 \end{aligned}$$

For element 2,

$$\begin{aligned}
 w^2(x) &= \sum_{j=1}^4 W_j \psi_j(x) \\
 &= W_3 \psi_1 + W_4 \psi_2 + 0 \times \psi_3 + 0 \times \psi_4 \\
 &= 2.083 \times 10^{-6} (1 - 3\bar{x}^2 + 2\bar{x}^3) + 1.042 \times 10^{-6} (\bar{x} - 2\bar{x}^2 + \bar{x}^3)
 \end{aligned}$$

Equation for Element 2 (using global x): Replace $\bar{x}$ with $(x - 1)$:

$$W_2(x) = \frac{25}{24 \times 10^6} [2 - (x - 1) - 4(x - 1)^2 + 3(x - 1)^3] \quad \text{for } 1 \leq x \leq 2$$



The solution code is as follow:

```

clear
% --- Data Setup ---
h = 1; q0 = 100; EI = 1e6;
syms x;

% Local Stiffness Matrix for an Euler-Bernoulli element
k= (EI/h^3) * [12, -6*h, -12, -6*h;

```

```

        -6*h, 4*h^2, 6*h, 2*h^2;
        -12, 6*h, 12, 6*h;
        -6*h, 2*h^2, 6*h, 4*h^2];

% Local Force Vector (Uniform load q0)
f1 = (q0*h/2) * [1; -h/6; 1; h/6];
f2= [0;0;0;0];
% --- Assembly & Boundary Conditions ---
% Since Node 1 (fixed) and Node 3 (fixed) have zero displacement,
% we only solve for the free degrees of freedom at Node 2 (index 3 and 4).
K_sub = k(3:4, 3:4) + k(1:2, 1:2); % Overlap at Node 2
F_sub = f1(3:4) + f2(1:2);         % Combined force at Node 2

% Solve for Node 2 displacements [w2; theta2]
W = K_sub \ F_sub
w2 = W(1); theta2 = W(2);

% --- Define Piecewise Solutions ---
% Element 1 (x from 0 to h) - Node 1 is [0,0], Node 2 is [w2, theta2]
N3 = (3*(x/h)^2 - 2*(x/h)^3);
N4 = x*((x/h)^2 - (x/h));
wx1 = N3*w2 + N4*theta2;

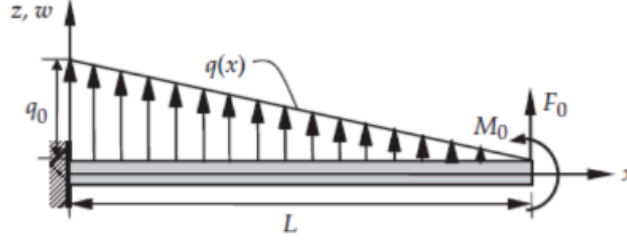
% Element 2 (x from h to 2h) - Node 2 is [w2, theta2], Node 3 is [0,0]
xi = x - h;
N1_e2 = 1 - 3*(xi/h)^2 + 2*(xi/h)^3;
N2_e2 = xi*(1 - xi/h)^2;
wx2 = N1_e2*w2 + N2_e2*theta2;

% --- Plotting ---
figure;
fplot(wx1, [0, h], 'b', 'LineWidth', 2); hold on;
fplot(wx2, [h, 2*h], 'r--', 'LineWidth', 2);
grid on; xlabel('x (m)'); ylabel('Deflection w (m)');
title('Fixed-Fixed Beam Deflection (2 Elements)');
legend('Element 1', 'Element 2');

```

2 Question 2

Use **two beam elements** to solve for the transverse displacement $W(x)$ of the following



Given:

$$\begin{aligned}
 E &= 100 \text{ GPa} \\
 L &= 1 \text{ m} \\
 t &= 0.1 \text{ m} \quad (\text{Square Cross Section}) \\
 F_0 &= 100 \text{ N} \\
 M_0 &= -10 \text{ N} \cdot \text{m} \quad (\text{J.N. Reddy Convention}) \\
 q(x) &= -100x + 100
 \end{aligned}$$

First, calculate the bending stiffness EI and the element constant.

- Moment of Inertia (I): For a square cross-section with side $t = 0.1 \text{ m}$:

$$I = \frac{t^4}{12} = \frac{(0.1)^4}{12} = \frac{0.0001}{12} = 8.333 \times 10^{-6} \text{ m}^4$$

- Bending Stiffness (EI): Given $E = 100 \text{ GPa} = 100 \times 10^9 \text{ Pa}$:

$$EI = (100 \times 10^9) \times \left(\frac{10^{-4}}{12} \right) = \frac{10^7}{12} \approx 833,333.33 \text{ N} \cdot \text{m}^2$$

- Element Length (h_e): Two elements for $L = 1 \text{ m}$; $h_e = \frac{L}{2} = 0.5 \text{ m}$
- Stiffness Coefficient (C): This constant appears in the stiffness matrix:

$$C = \frac{2EI}{h_e^3} = \frac{2 \times 10^7 / 12}{(0.5)^3} = \frac{2 \times 10^7}{12 \times 0.125} = \frac{2 \times 10^7}{1.5} \approx 13.3334 \times 10^6 \text{ N/m}$$

2.1 I. Find the element wise stiffness matrix and force vectors.

We define the local stiffness matrix and force vector for a single element e .

Let $L_e = 0.5$, $E = 100 \text{ GPa}$, $I = 8.333 \times 10^{-6} \text{ m}^4$.

Still using the formula provided previously:

$$k^e = \frac{2EI}{L_e^3} \begin{bmatrix} 6 & -3L_e & -6 & -3L_e \\ -3L_e & 2L_e^2 & 3L_e & L_e^2 \\ -6 & 3L_e & 6 & 3L_e \\ -3L_e & L_e^2 & 3L_e & 2L_e^2 \end{bmatrix} \quad (6)$$

Substitute C and L into Equation 6,

$$\begin{aligned}
 k^e &= C \begin{bmatrix} 6 & -3(0.5) & -6 & -3(0.5) \\ -3(0.5) & 2(0.5)^2 & 3(0.5) & (0.5)^2 \\ -6 & 3(0.5) & 6 & 3(0.5) \\ -3(0.5) & (0.5)^2 & 3(0.5) & 2(0.5)^2 \end{bmatrix} \\
 &= 13.3334 \times 10^6 \begin{bmatrix} 6 & -1.5 & -6 & -1.5 \\ -1.5 & 0.5 & 1.5 & 0.25 \\ -6 & 1.5 & 6 & 1.5 \\ -1.5 & 0.25 & 1.5 & 0.5 \end{bmatrix}
 \end{aligned}$$

It applies to both Element 1 and Element 2 since they have identical geometry and material.

B. Distributed Load Vector (f_{dist}^e)

For the distributed load $q(x) = -100x + 100$:

- Element 1 (e^1): $x \in [0, 0.5]$. Load varies linearly from $q_a = 100$ to $q_b = 50$.
- Element 2 (e^2): $x \in [0.5, 1.0]$. Load varies linearly from $q_a = 50$ to $q_b = 0$.

We have the shape functions as follows:

$$\begin{aligned}
 \psi_1^e(x) &= 1 - 3 \left(\frac{\bar{x}}{L_e} \right)^2 + 2 \left(\frac{\bar{x}}{L_e} \right)^3 \\
 \psi_2^e(x) &= -\bar{x} \left(1 - \frac{\bar{x}}{L_e} \right)^2 \\
 \psi_3^e(x) &= 3 \left(\frac{\bar{x}}{L_e} \right)^2 - 2 \left(\frac{\bar{x}}{L_e} \right)^3 \\
 \psi_4^e(x) &= -\bar{x} \left(-\frac{\bar{x}}{L_e} + \left(\frac{\bar{x}}{L_e} \right)^2 \right)
 \end{aligned}$$

For a linear load varying from q_a (left) to q_b (right) over length L_e , the consistent force vector is:

For element 1:

$$f_1^1 = \int_0^{0.5} \psi_1^1(x) q(x) dx = \int_0^{0.5} \left(1 - 3 \left(\frac{x}{0.5} \right)^2 + 2 \left(\frac{x}{0.5} \right)^3 \right) (-100x + 100) dx = 21.25$$

$$f_2^1 = \int_0^{0.5} \psi_2^1(x) q(x) dx = \int_0^{0.5} \left(-x \left(1 - \frac{x}{0.5} \right)^2 \right) (-100x + 100) dx = -\frac{5}{3}$$

$$f_3^1 = \int_0^{0.5} \psi_3^1(x) q(x) dx = \int_0^{0.5} \left(3 \left(\frac{x}{0.5} \right)^2 - 2 \left(\frac{x}{0.5} \right)^3 \right) (-100x + 100) dx = 16.25$$

$$f_4^1 = \int_0^{0.5} \psi_4^1(x) q(x) dx = \int_0^{0.5} \left(-x \left(-\frac{x}{0.5} + \left(\frac{x}{0.5} \right)^2 \right) \right) (-100x + 100) dx = \frac{35}{24}$$

For element 2:

$$\begin{aligned}
f_1^2 &= \int_{0.5}^1 \psi_1^2(x) q(x) dx = \int_{0.5}^1 \left(1 - 3 \left(\frac{x-0.5}{0.5} \right)^2 + 2 \left(\frac{x-0.5}{0.5} \right)^3 \right) (-100x + 100) dx = 8.75 \\
f_2^2 &= \int_{0.5}^1 \psi_2^2(x) q(x) dx = \int_{0.5}^1 \left(-(x-0.5) \left(1 - \frac{x-0.5}{0.5} \right)^2 \right) (-100x + 100) dx = -0.625 \\
f_3^2 &= \int_{0.5}^1 \psi_3^2(x) q(x) dx = \int_{0.5}^1 \left(3 \left(\frac{x-0.5}{0.5} \right)^2 - 2 \left(\frac{x-0.5}{0.5} \right)^3 \right) (-100x + 100) dx = 3.75 \\
f_4^2 &= \int_{0.5}^1 \psi_4^2(x) q(x) dx = \int_{0.5}^1 \left(-(x-0.5) \left(-\frac{x-0.5}{0.5} + \left(\frac{x-0.5}{0.5} \right)^2 \right) \right) (-100x + 100) dx = -\frac{5}{12}
\end{aligned}$$

2.2 II. Assemble the system.

For this beam, we use two beam elements to discretize and assemble it.

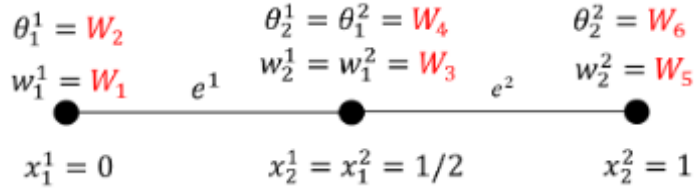


Figure 2: Symbol numbering system

Each beam element has 4 degrees of freedom. Assign a coordinate system as follows:

- x_n^e : Position of node n for element e
- w_n^e : Local Transverse displacement of node n for element e
- θ_n : Local Slope of node n for element e
- W : Global degree of freedom

The global system has 3 nodes and 6 Degrees of Freedom: $W_1, \theta_1, W_2, \theta_2, W_3, \theta_3$.

Since this system has 6 independent degrees of freedom, the global system will be:

$$\{K\}_{6 \times 6} \{W\}_{6 \times 6} = \{f\}_{6 \times 6}$$

Assemble Equation 1 and Equation 2,

We assemble the global system of size 6×6 to solve for the vector $\mathbf{W} = [W_1, W_2, W_3, W_4, W_5, W_6]^T$.

A. Element Mapping

Element 1: Corresponds to global indices [1, 2, 3, 4].

$$K^{(1)} = 13.3334 \times 10^6 \begin{bmatrix} 6 & -1.5 & -6 & -1.5 \\ -1.5 & 0.5 & 1.5 & 0.25 \\ -6 & 1.5 & 6 & 1.5 \\ -1.5 & 0.25 & 1.5 & 0.5 \end{bmatrix}, \quad F^{(1)} = \begin{bmatrix} 21.25 \\ -\frac{5}{3} \\ 16.25 \\ \frac{35}{24} \end{bmatrix}$$

Element 2: Corresponds to global indices [3, 4, 5, 6].

$$K^{(2)} = 13.3334 \times 10^6 \begin{bmatrix} 6 & -1.5 & -6 & -1.5 \\ -1.5 & 0.5 & 1.5 & 0.25 \\ -6 & 1.5 & 6 & 1.5 \\ -1.5 & 0.25 & 1.5 & 0.5 \end{bmatrix}, \quad F^{(2)} = \begin{bmatrix} 8.75 \\ -0.625 \\ 3.75 \\ \frac{5}{12} \end{bmatrix}$$

B. Global Stiffness Matrix

Overlap occurs at indices 3 and 4 (Node 2).

$$\begin{aligned} K_{global} &= \begin{bmatrix} k_{11}^1 & k_{12}^1 & k_{13}^1 & k_{14}^1 & 0 & 0 \\ k_{21}^1 & k_{22}^1 & k_{23}^1 & k_{24}^1 & 0 & 0 \\ k_{31}^1 & k_{32}^1 & \mathbf{k_{33}^1 + k_{11}^2} & \mathbf{k_{34}^1 + k_{12}^2} & k_{13}^2 & k_{14}^2 \\ k_{41}^1 & k_{42}^1 & \mathbf{k_{43}^1 + k_{21}^2} & \mathbf{k_{44}^1 + k_{22}^2} & k_{23}^2 & k_{24}^2 \\ 0 & 0 & k_{31}^2 & k_{32}^2 & k_{33}^2 & k_{34}^2 \\ 0 & 0 & k_{41}^2 & k_{42}^2 & k_{43}^2 & k_{44}^2 \end{bmatrix} \\ &= 13.3334 \times 10^6 \begin{bmatrix} 6 & -1.5 & -6 & -1.5 & 0 & 0 \\ -1.5 & 0.5 & 1.5 & 0.25 & 0 & 0 \\ -6 & 1.5 & 12 & 0 & -6 & -1.5 \\ -1.5 & 0.25 & 0 & 1 & 1.5 & 0.25 \\ 0 & 0 & -6 & 1.5 & 6 & 1.5 \\ 0 & 0 & -1.5 & 0.25 & 1.5 & 0.5 \end{bmatrix} \end{aligned}$$

C. Global Force Vector

Includes distributed loads plus the point loads (F_0, M_0) at Node 3 (indices 5 and 6).

$$\begin{aligned} F_{global} &= \begin{bmatrix} f_1^1 \\ f_2^1 \\ f_3^1 + f_1^2 \\ f_4^1 + f_2^2 \\ f_3^2 + F_0 \\ f_4^2 + M_0 \end{bmatrix} \\ &= \begin{bmatrix} 21.25 \\ -\frac{5}{3} \\ 16.25 + 8.75 \\ \frac{35}{24} - 0.625 \\ 3.75 + 100 \\ \frac{5}{12} - 10 \end{bmatrix} \end{aligned}$$

Where:

- $F_0 = 100$
- $M_0 = -10$ (Ensure this sign is consistent with your clockwise positive convention).

2.3 III. Implement the boundary conditions.

The beam is fixed at the left end (Node 1). At $x = 0$: $W_1 = 0$ and $W_2 = 0$.

This means we remove the **1st and 2nd rows** and the **1st and 2nd columns** from the system. We are left with a 4×4 system solving for indices 3, 4, 5, 6.

So, we can reduce System:

$$\mathbf{K}_{red} = 13.3334 \times 10^6 \begin{bmatrix} 12 & 0 & -6 & -1.5 \\ 0 & 1 & 1.5 & 0.25 \\ -6 & 1.5 & 6 & 1.5 \\ -1.5 & 0.25 & 1.5 & 0.5 \end{bmatrix}$$

$$\mathbf{F}_{red} = \begin{bmatrix} 16.25 + 8.75 \\ \frac{35}{24} - 0.625 \\ 3.75 + 100 \\ \frac{5}{12} - 10 \end{bmatrix}$$

Final Solution:

$$\mathbf{W}_{red} = \mathbf{K}_{red}^{-1} \mathbf{F}_{red}$$

2.4 IV. Solve for the nodal values and plot $W(x)$

Given: $h_e = 0.5$ m, $EI = 833,333.33$ N · m², $F_0 = 100$ N, $M_0 = -10$ N · m.

Solve $\mathbf{W}_{red} = \mathbf{K}_{red}^{-1} \mathbf{F}_{red}$:

W1 (w1)	=	0.000000e+00	m
W2 (th1)	=	0.000000e+00	rad
W3 (w2)	=	1.553125e-05	m
W4 (th2)	=	-5.568750e-05	rad
W5 (w3)	=	5.000000e-05	m
W6 (th3)	=	-7.700000e-05	rad

That is:

$$W_3 = 1.5531 \times 10^{-5} \text{ m}, \quad W_4 = -5.5688 \times 10^{-5} \text{ rad}$$

$$W_5 = 5.0000 \times 10^{-5} \text{ m}, \quad W_6 = -7.7000 \times 10^{-5} \text{ rad}$$

2.4.1 Reconstruct $w(x)$ using shape functions

To construct the full solution:

$$w^e(\bar{x}) = \sum_{j=1}^4 W_j^e \psi_j^e(\bar{x})$$

where the shape functions (with $L_e = 0.5$) are:

$$\psi_1(\bar{x}) = 1 - 12\bar{x}^2 + 16\bar{x}^3$$

$$\psi_2(\bar{x}) = -\bar{x}(1 - 2\bar{x})^2$$

$$\psi_3(\bar{x}) = 12\bar{x}^2 - 16\bar{x}^3$$

$$\psi_4(\bar{x}) = 2\bar{x}^2 - 4\bar{x}^3$$

Element 1 ($0 \leq x \leq 0.5$, $\bar{x} = x$): Since $W_1 = W_2 = 0$,

$$\begin{aligned} w^1(x) &= W_3 \psi_3(x) + W_4 \psi_4(x) \\ &= 1.5531 \times 10^{-5} (12x^2 - 16x^3) + (-5.5688 \times 10^{-5}) (2x^2 - 4x^3) \end{aligned}$$

Element 2 ($0.5 \leq x \leq 1.0$, $\bar{x} = x - 0.5$):

$$\begin{aligned} w^2(\bar{x}) &= W_3\psi_1(\bar{x}) + W_4\psi_2(\bar{x}) + W_5\psi_3(\bar{x}) + W_6\psi_4(\bar{x}) \\ &= 1.5531 \times 10^{-5}(1 - 12\bar{x}^2 + 16\bar{x}^3) \\ &\quad + (-5.5688 \times 10^{-5})(-\bar{x}(1 - 2\bar{x})^2) \\ &\quad + 5.0000 \times 10^{-5}(12\bar{x}^2 - 16\bar{x}^3) \\ &\quad + (-7.7000 \times 10^{-5})(2\bar{x}^2 - 4\bar{x}^3) \end{aligned}$$

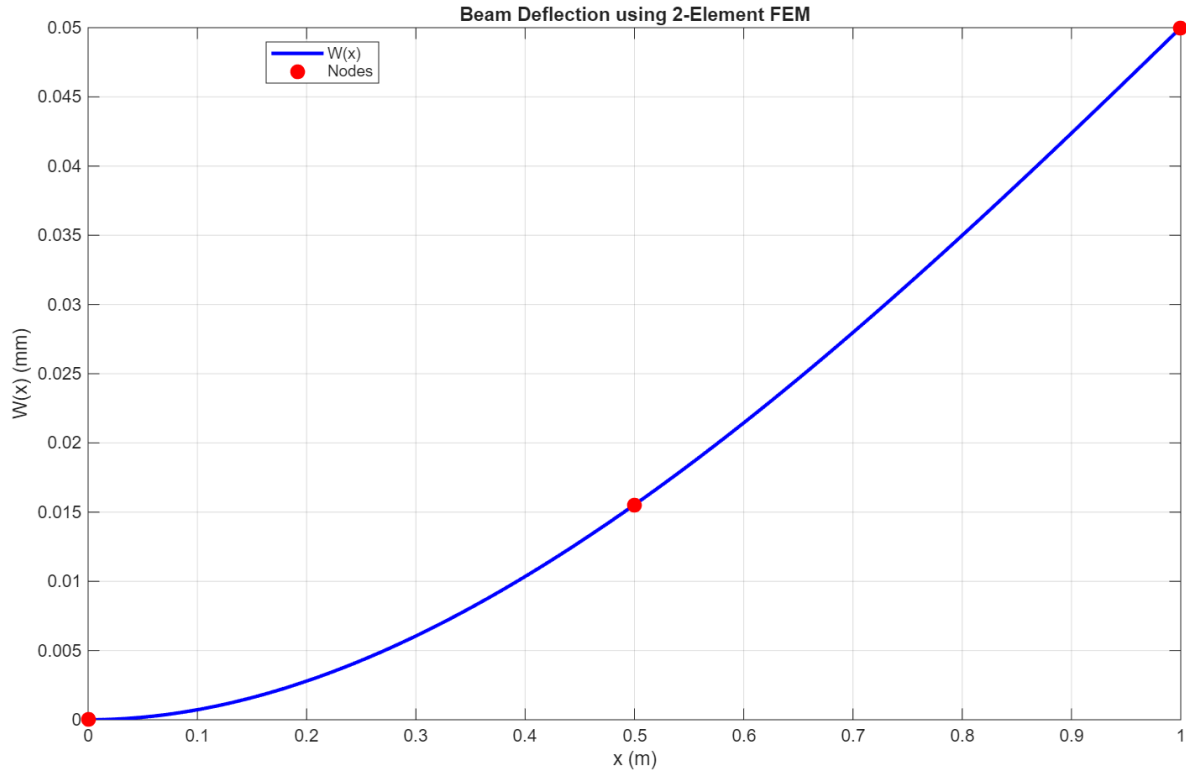


Figure 3: Deflection of Cantilever Beam

The solution code is as follows:

```
%% CIV E 665 - Assignment 5, Question 2
% Two-element Euler-Bernoulli beam FEM
% Cantilever beam: fixed at left end, free at right end
% q(x) = -100x + 100, FO = 100 N, MO = -10 N·m

clear; clc; close all;

%% Given data
E = 100e9; % Young's modulus [Pa]
L = 1; % Total beam length [m]
t = 0.1; % Square cross-section side [m]
FO = 100; % Point force at free end [N]
MO = -10; % Moment at free end [N·m] (J.N. Reddy convention)
```

```

nElem = 2;
Le = L / nElem;    % Element length = 0.5 m

%% Cross-section properties
I = t^4 / 12;      % Moment of inertia
EI = E * I;        % Bending stiffness
C = 2 * EI / Le^3; % Stiffness coefficient

fprintf('I = %.6e m^4\n', I);
fprintf('EI = %.4f N·m^2\n', EI);
fprintf('C = %.4e N/m\n', C);

%% Element stiffness matrix (same for both elements)
ke = C * [ 6,      -3*Le,  -6,      -3*Le;
          -3*Le,  2*Le^2,  3*Le,    Le^2;
          -6,     3*Le,   6,       3*Le;
          -3*Le,  Le^2,   3*Le,    2*Le^2];

%% Element force vectors from distributed load q(x) = -100x + 100
% Element 1: x in [0, 0.5]
syms x
Le_val = 0.5;

% Shape functions for element 1 (xbar = x)
psi1_e1 = 1 - 3*(x/Le_val)^2 + 2*(x/Le_val)^3;
psi2_e1 = -x*(1 - x/Le_val)^2;
psi3_e1 = 3*(x/Le_val)^2 - 2*(x/Le_val)^3;
psi4_e1 = -x*(-x/Le_val + (x/Le_val)^2);

q = -100*x + 100;

f1 = double([int(psi1_e1 * q, x, 0, 0.5);
             int(psi2_e1 * q, x, 0, 0.5);
             int(psi3_e1 * q, x, 0, 0.5);
             int(psi4_e1 * q, x, 0, 0.5)]);

% Element 2: x in [0.5, 1.0], xbar = x - 0.5
xbar = x - 0.5;
psi1_e2 = 1 - 3*(xbar/Le_val)^2 + 2*(xbar/Le_val)^3;
psi2_e2 = -xbar*(1 - xbar/Le_val)^2;
psi3_e2 = 3*(xbar/Le_val)^2 - 2*(xbar/Le_val)^3;
psi4_e2 = -xbar*(-xbar/Le_val + (xbar/Le_val)^2);

f2 = double([int(psi1_e2 * q, x, 0.5, 1.0);
             int(psi2_e2 * q, x, 0.5, 1.0);
             int(psi3_e2 * q, x, 0.5, 1.0);
             int(psi4_e2 * q, x, 0.5, 1.0)]);

fprintf('\nElement 1 force vector:\n');
disp(f1);
fprintf('Element 2 force vector:\n');

```

```

disp(f2);

%% Assemble global stiffness matrix (6x6)
K = zeros(6, 6);
K(1:4, 1:4) = K(1:4, 1:4) + ke;    % Element 1 -> DOFs 1-4
K(3:6, 3:6) = K(3:6, 3:6) + ke;    % Element 2 -> DOFs 3-6

%% Assemble global force vector (6x1)
F = zeros(6, 1);
F(1:4) = F(1:4) + f1;              % Element 1
F(3:6) = F(3:6) + f2;              % Element 2

% Add point loads at Node 3 (free end): F0 and M0
F(5) = F(5) + F0;
F(6) = F(6) + M0;

fprintf('Global force vector F:\n');
disp(F);

%% Apply boundary conditions: fixed at Node 1 => W1=0, W2=0
% Reduce system: remove DOFs 1 and 2
freeDOFs = [3, 4, 5, 6];
K_red = K(freeDOFs, freeDOFs);
F_red = F(freeDOFs);

fprintf('Reduced stiffness matrix K_red:\n');
disp(K_red);
fprintf('Reduced force vector F_red:\n');
disp(F_red);

%% Solve
W_red = K_red \ F_red;

% Full solution vector
W = zeros(6, 1);
W(freeDOFs) = W_red;

fprintf('Nodal solution vector W:\n');
fprintf('  W1 (w at node 1) = %.6e m\n', W(1));
fprintf('  W2 (theta at node 1) = %.6e rad\n', W(2));
fprintf('  W3 (w at node 2) = %.6e m\n', W(3));
fprintf('  W4 (theta at node 2) = %.6e rad\n', W(4));
fprintf('  W5 (w at node 3) = %.6e m\n', W(5));
fprintf('  W6 (theta at node 3) = %.6e rad\n', W(6));

%% Compute reactions at fixed end
R = K * W - F;
% Only non-zero at constrained DOFs (1 and 2)
fprintf('\nReactions at fixed end:\n');
fprintf('  R1 (shear force) = %.4f N\n', -R(1));
fprintf('  R2 (moment)      = %.4f N·m\n', -R(2));

```

```

%% Plot W(x)
nPts = 100;

% Element 1: x in [0, 0.5]
x1 = linspace(0, Le, nPts);
xbar1 = x1 / Le;
N1 = [1 - 3*xbar1.^2 + 2*xbar1.^3;
      -x1.*(1 - xbar1).^2;
      3*xbar1.^2 - 2*xbar1.^3;
      -x1.*(-xbar1 + xbar1.^2)];
w1 = W(1:4)' * N1;

% Element 2: x in [0.5, 1.0]
x2 = linspace(Le, 2*Le, nPts);
xbar2 = (x2 - Le) / Le;
xb = x2 - Le; % local coordinate
N2 = [1 - 3*xbar2.^2 + 2*xbar2.^3;
      -xb.*(1 - xbar2).^2;
      3*xbar2.^2 - 2*xbar2.^3;
      -xb.*(-xbar2 + xbar2.^2)];
w2 = W(3:6)' * N2;

% Combine and plot
X = [x1, x2];
W_plot = [w1, w2];

figure;
plot(X, W_plot * 1e3, 'b-', 'LineWidth', 2);
xlabel('x (m)');
ylabel('W(x) (mm)');
title('Beam Deflection using 2-Element FEM');
grid on;

% Mark nodes
hold on;
nodes_x = [0, Le, 2*Le];
nodes_w = [W(1), W(3), W(5)] * 1e3;
plot(nodes_x, nodes_w, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
legend('W(x)', 'Nodes', 'Location', 'best');
hold off;

```